SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement –**

Intelligent Public Transport Crowding & Route Prediction

- **Theme-Transportation & Logistics**
- **PS Category- Software**
- **Team Name - Medha**

Medha

TRANSITPULSE

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Proposed Solution

- Per-stop crowding & delay prediction
- Real-time formulas: time, weather, disruptions
- Cascade model: disruption spreads with distance
- Auto route-finding + best option ranking
- Naive vs recommended comparison
- XGBoost model running alongside rule engine

How it addresses the problem

- Removes commuter guesswork
- Early cascade warning for authorities

Innovation & uniqueness

- Shows how disruptions spread, not just current status
- Naive-vs-optimized comparison makes benefit tangible
- Lightweight, explainable core — easy to upgrade to ML

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TECHNICAL APPROACH

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Technologies used

- Python (Pandas, NumPy) — data & scoring engine
- XGBoost – trained crowding model, validated on held-out test data
- FastAPI — backend serving predictions
- React + Recharts — live dashboard
- GTFS feed — real route/stop data
- SQLite — lightweight data store
- Open-Meteo – live weather intensity, no API key required
- BMTC GTFS (Vonter/bmtc-gtfs) – real Bengaluru route/stop network

Methodology — process flow

Data ingestion (historical + live) → Crowding/delay scoring engine → Cascade propagation → Recommendation engine → FastAPI backend → React dashboard

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FEASIBILITY AND VIABILITY

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Feasibility

•Uses open GTFS standard — no proprietary data needed
•No training data or GPU required
•Phased rollout: one corridor first, then scale

Potential challenges & risks

•Live data feeds vary by transit agency
•Accuracy depends on historical data quality
•Needs integration with an existing transit app

Strategies to overcome them

•Fallback: static schedule + crowdsourced reports
•Phased data-sharing deal with transit authority
•Ship as API/plugin, not a standalone app

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IMPACT AND BENEFITS

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Potential impact on target audience

- Commuters get clear, actionable route guidance
- Authorities get early warning of disruptions

Benefits

- Social: less commute stress, safer boarding
- Economic: better fleet use, less time lost
- Environmental: shift from private vehicles to transit

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RESEARCH AND REFERENCES

•GTFS standard — gtfs.org

•Open transit datasets — data.gov.in

•Internal causal-graph model, adapted from network-fault prediction

•Literature on transit crowding & delay propagation

• XGBoost — trained model docs